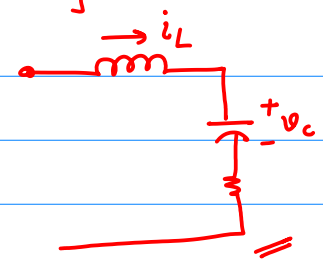
$\boxed{L3}$

$$\dot{x} = Ax + Bu \longrightarrow (1) \qquad T^{-1}AT = \begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix}$$



Let $x(t) = T\hat{x}(t)$

$$T\dot{\hat{x}}(t) = AT\hat{x}(t) + Bu$$

$$\Rightarrow \dot{\hat{x}}(t) = \underbrace{T^{-1}AT}\,\hat{x}(t) + \underbrace{T^{-1}B}u$$

$$\dot{\hat{x}}(t) = \left[\begin{array}{c|c} \overset{r\times r}{\tilde{A}_{11}} & \tilde{A}_{12} \\ \hline 0 & \underset{(n-r)\times(n-r)}{\tilde{A}_{22}} \end{array}\right]\hat{x}(t) + \begin{bmatrix} \overset{r\times m}{\tilde{B}_1} \\ 0 \end{bmatrix} u$$

$$^{*}T = [\underbrace{b_1\ b_2\ b_3}_{\text{Basis using } \mathcal{C}_{AB}}\ v_1\ v_2]_{n\times n}$$

$$\frac{d}{dt}\begin{bmatrix} \hat{x}_1(t) \\ \hat{x}_2(t) \end{bmatrix} = \begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix}\begin{bmatrix} \hat{x}_1(t) \\ \hat{x}_2(t) \end{bmatrix} + \begin{bmatrix} \tilde{B}_1 \\ 0 \end{bmatrix} u \qquad \hat{x} = \begin{bmatrix} v_c \\ i_L \\ v_c \end{bmatrix}$$

$$\boxed{\frac{d}{dt}\hat{x}_2(t) = \underline{\underline{\tilde{A}_{22}}}\,\hat{x}_2(t)}$$

Autonomous system

$$\mathbb{R}^n = \mathcal{X}_{con} \oplus \mathcal{X}/\mathcal{X}_{con}$$

<u>Autonomous system:</u> $\boxed{\dot{p}(t) = Mp(t)} \Rightarrow \underline{\underline{p(t)}} = e^{Mt}p(0)$

$M \in \mathbb{R}^{n\times n}$

<u>Case I:</u> M is diagonalizable. $\exists\, U \in \mathbb{R}^{n\times n}$ such that

$$U^{-1}\underline{\underline{M}}U = \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}}_{\doteq D} \Rightarrow M = UDU^{-1}$$

(How do we get U?) Eigenvectors

<u>Why?</u> Ex

$$e^{Mt} = e^{UDU^{-1}t}$$

$$= \left[UU^{-1} + (UDU^{-1})t + (UDU^{-1})^2\frac{t^2}{\lfloor 2} + \cdots\right]$$

$$= U\left[I + Dt + D^2t^2/\lfloor 2 + \cdots\right]U^{-1} = Ue^{Dt}U^{-1}$$

$$\therefore\ e^{Mt} = \underline{U}\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & e^{\lambda_n t} \end{bmatrix}\underline{U^{-1}}$$

$$\overset{U}{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}\begin{bmatrix} e^t & \\ & e^{-t} \end{bmatrix}\overset{U^{-1}}{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}$$

$$= \begin{bmatrix} e^t & e^t - e^{-t} \\ 0 & e^{-t} \end{bmatrix}$$

Nilpotent $N^k = 0$

Case II: The matrix M has one Jordan block. $\exists W \in \mathbb{R}^{n\times n}$

$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $N^2 = 0$

$N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $N^2, N^3 = 0$

$$W^{-1} M W = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & & \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{bmatrix}$$

$$= \lambda I + N, \qquad N := \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$\Rightarrow M = W(\lambda I + N)W^{-1}$$

$$e^{Mt} = W e^{(\lambda I + N)t} W^{-1} = W e^{\lambda t} \cdot e^{Nt} W^{-1}$$

$$= W e^{\lambda t}\left[I + Nt + \frac{N^2 t^2}{\lfloor 2} + \cdots \frac{N^{n-1} t^{n-1}}{\lfloor n-1}\right] W^{-1}$$

$e^{A+B} = e^A \cdot e^B$

Infim Infim

$$= W\left(e^{\lambda t}\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} + e^{\lambda t}\begin{bmatrix} 0 & t & 0 & \cdots & 0 \\ 0 & 0 & t & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} + e^{\lambda t}\begin{bmatrix} 0 & 0 & t^2/\lfloor 2 & 0 & \cdots & 0 \\ 0 & 0 & 0 & t^2/\lfloor 2 & \cdots & 0 \\ \vdots & & & & & \\ \cdot & \cdot & & & & \cdot \end{bmatrix}\right) W^{-1}$$

$AB = BA$ Commute

$$= W e^{\lambda t}\begin{bmatrix} 1 & t & t^2/\lfloor 2 & \cdots & t^{n-1}/\lfloor n-1 \\ 0 & 1 & t & \cdots & t^{n-2}/\lfloor n-2 \\ \vdots & \vdots & \ddots & & \\ 0 & 0 & \cdots & \cdots & 1 \end{bmatrix} W^{-1}$$

$\sigma(M) = \{\lambda_1, \lambda_2 \cdots \lambda_r\}$

Case III: $\exists V \in \mathbb{R}^{n\times n}$,

$$V^{-1} M V = \begin{bmatrix} \lambda_1 & 1 & & & & & & \\ 0 & \lambda_1 & & & & & & \\ & & \lambda_2 & & & & & \\ & & & \lambda_3 & & & & \\ & & & & \ddots & & & \\ & & & & & \lambda_r & 1 & \\ & & & & & & \lambda_r & 1 \\ & & & & & & & \lambda_r \end{bmatrix}$$

Algebraic multiplicity,

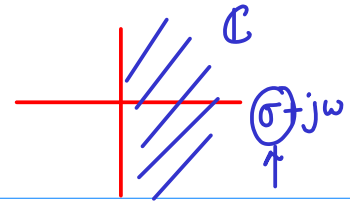
Geometric multiplicity

$$\begin{bmatrix} \lambda_1 & 1 & \\ & \lambda_1 & \\ & & \lambda_1 \end{bmatrix}$$

$$e^{Mt} = V \begin{bmatrix} e^{\lambda_1 t} & te^{\lambda_1 t} & & & & & & \\ 0 & e^{\lambda_1 t} & & & & & & \\ & & e^{\lambda_2 t} & & & & & \\ & & & e^{\lambda_3 t} & & & & \\ & & & & \ddots & & & \\ & & & & & e^{\lambda_r t} & te^{\lambda_r t} & t^2 e^{\lambda_r t}/2 \\ & & & & & & e^{\lambda_r t} & te^{\lambda_r t} \\ & & & & & & & e^{\lambda_r t} \end{bmatrix} V^{-1}$$

$e^{Mt} \rightarrow e^{\lambda_1 t}, e^{\lambda_2 t} \ldots$

$\sigma(A) \notin \mathbb{C}^-$



$e^{\lambda_1 t} = e^{(\sigma + j\omega)t}$

$= e^{\sigma t} e^{j\omega t}$

$\sigma t \rightarrow$ blow up

$e^{j\omega t} \rightarrow 1$

$\cos\omega t + j\sin\omega t$

$\lim_{t\to\infty} e^{Mt} \uparrow$ blows up

$\boxed{\sigma(M) \in \mathbb{C}^-}$

$\boxed{\lim_{t\to\infty} e^{Mt} = 0}$

Internal stability

$\boxed{\lim_{t\to\infty} x(t) = 0}$

Th^{m}: An autonomous system is internally stable if and only M is Hurwitz.

$\downarrow$

$\sigma(M) \in \mathbb{C}^-$

Stability

$\dot{x} = Ax \rightarrow$ internally stable.

$\dot{x} = Ax + Bu$

BIBO stability

$$\boxed{\frac{d}{dt}\hat{x}_2(t) = \tilde{A}_{22}\hat{x}_2(t)}$$

$\sigma(\tilde{A}_{22}) \subseteq \mathbb{C}^- \Rightarrow$ System is stabilizable.

$\underbrace{\text{Controllability}}_{u}$ $\underbrace{\text{Stabilizability}}_{\text{Uncontrollable}}$ $\underbrace{\text{Autonomous}}_{x}$

x_0

$\mathcal{X} = \mathcal{X}_{cont} \oplus \mathcal{X}/\mathcal{X}_{cont}$

$$\begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix} \rightarrow \dot{\tilde{x}}_2 = \tilde{A}_{22}\tilde{x}_2$$